

LIZMAP WEB CLIENT

Web Geoportal for QGIS

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3Liz

- French company
- **Open Source** forever, check our GitHub <https://github.com/3liz/>
- Core contributors in QGIS, server side mainly
- PyQGIS development, Lizmap server hosting (SAAS), trainings about PyQGIS, PostGIS, Lizmap...

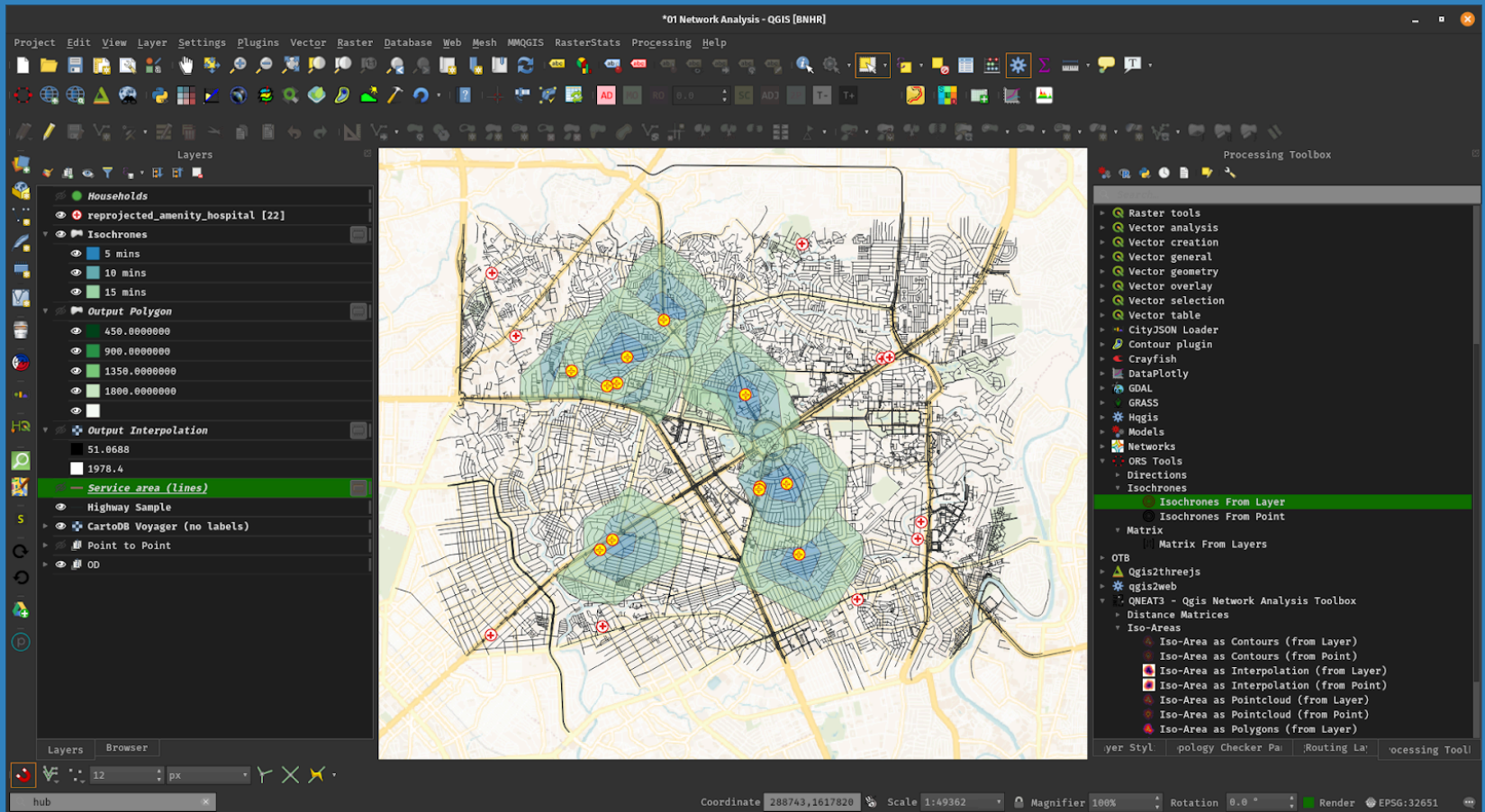


QGIS Desktop

We sometimes spend a lot of time configuring our project in QGIS Desktop 🕒

- Setting up layers and relations
- Layers symbology
- Build form to make it easy to add data
- PDF print layout
- ...

QGIS Desktop



Need to share your work ?

- We already use QGIS as our main GIS tool
- We **are used to it** !
- Then we need to publish a full-featured application to the web ?
- We **do not** want to spend **again** time to set up and configure web maps in complex admin panels, re-styling layers, forms etc.

One thing to know

- QGIS is **not only** a desktop application
- But a GIS server as well 🚀
- OGC certification since 2018 : WMS, WFS, WMTS...



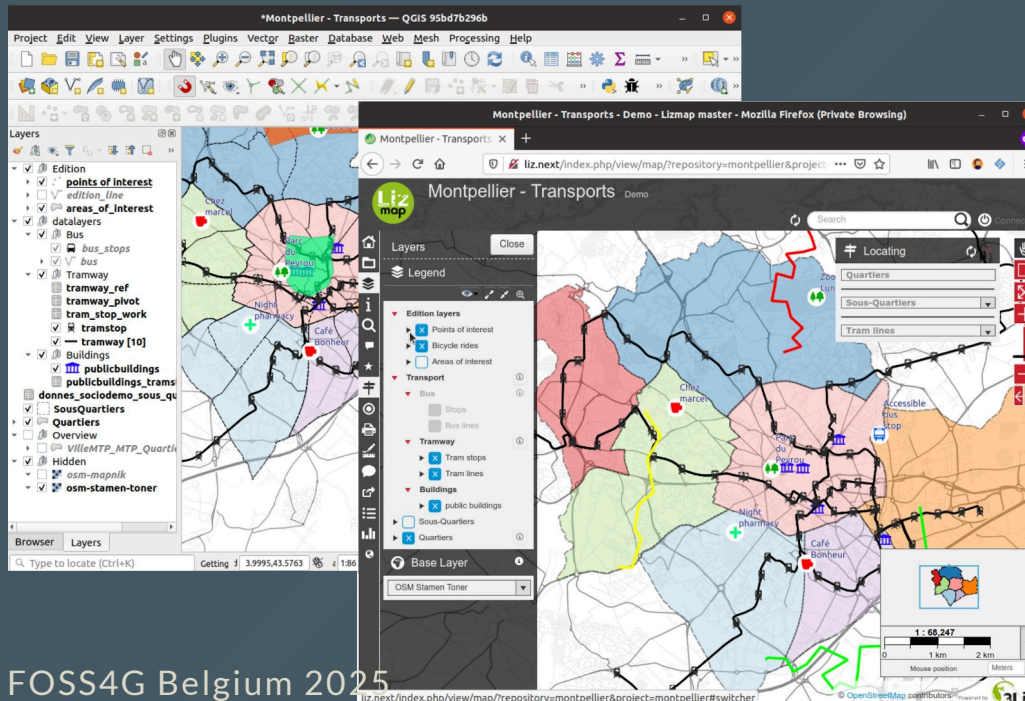
On top of QGIS Server

- Lizmap Web Client
- Prepare on QGIS desktop, deploy on Lizmap
- Github, open source : Mozilla Public Licence
- Web administration panel is **mainly** for authentication and authorization management (users and groups)
- All other configurations are done **within QGIS desktop**



How to

- Create a project with some layers
- Use the Lizmap plugin to configure some options specific for the web (extent, scales, tools available etc.)
- And upload on the Lizmap server
- You've got a web map based on the QGIS project



Use cases of Lizmap

Some examples to demonstrate
the main features

Map Catalog in Meylan

French city in the Alps
Publish thematic maps for citizens
Eye-candy, with simplicity

<https://geo.meylan.fr>

Equipements



meylan
Des services partagés

Aires de jeux

[Voir la carte](#)

[Description](#)



meylan
Des services partagés

Culture

[Voir la carte](#)

[Description](#)



meylan
Des services partagés

Petite enfance

[Voir la carte](#)

[Description](#)



meylan
Des services partagés

Scolaire et Péri-scolaire

[Voir la carte](#)

[Description](#)



meylan
Des services partagés

Sports

[Voir la carte](#)

[Description](#)



meylan
Des services partagés

Tous équipements

[Voir la carte](#)

[Description](#)

Popup

Fermer

Aire de jeux ouverte au public

ouvert au public



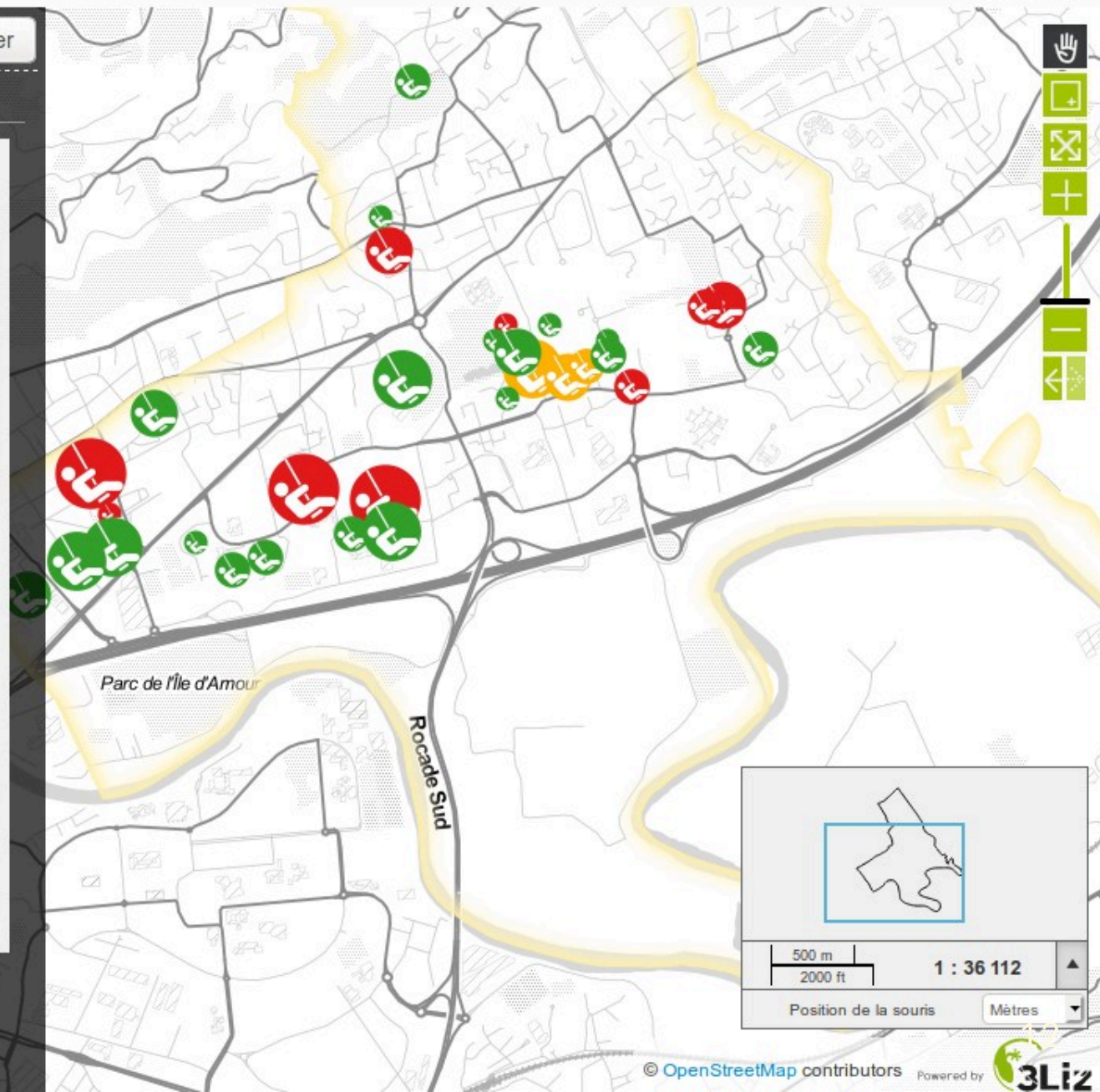
Nombre de Jeux : 3

Tranche d'âge autorisée : enfants entre 3-12 ans

Lieu : Parc des Saules

Quartier : Aiguinards

Informations complémentaires : Aires de jeux pour enfants. La plupart des aires de jeux sont ouvertes au public. Toutefois, certaines aires de jeux sont fermées au public (école, crèche...). D'autres, sont ouvertes en dehors du temps scolaire (semi-ouverte). Source : Ville de Meylan



Custom CSS themes

- <https://mapeo.calvados.fr/carto/>



Connexion

Inscription

légende



Changer le thème de la carte



- ☒  **Siège EPCI**
- ☒  **Mairies**
- ☒  **Mairies déléguées et associées**
- ☐  **Bureaux de vote**
- ☐  **Syndicats**
- ☒ **Limites administratives**
 - ☒  **Département**
 - ☐  **Circonscription législatives**
 - ☐  **EPCI**
 - ☒  **Canton**



Fond de carte



Plan IGN



Localisation



Commune



Canton



EPCI



Pointe de Barfleur

BAIE

SE



St-Lô

Normandie

Piers

Baignolles de l'Orne

Normandie

1 : 1 000 000

0 25 km 50 km

327811

6990891



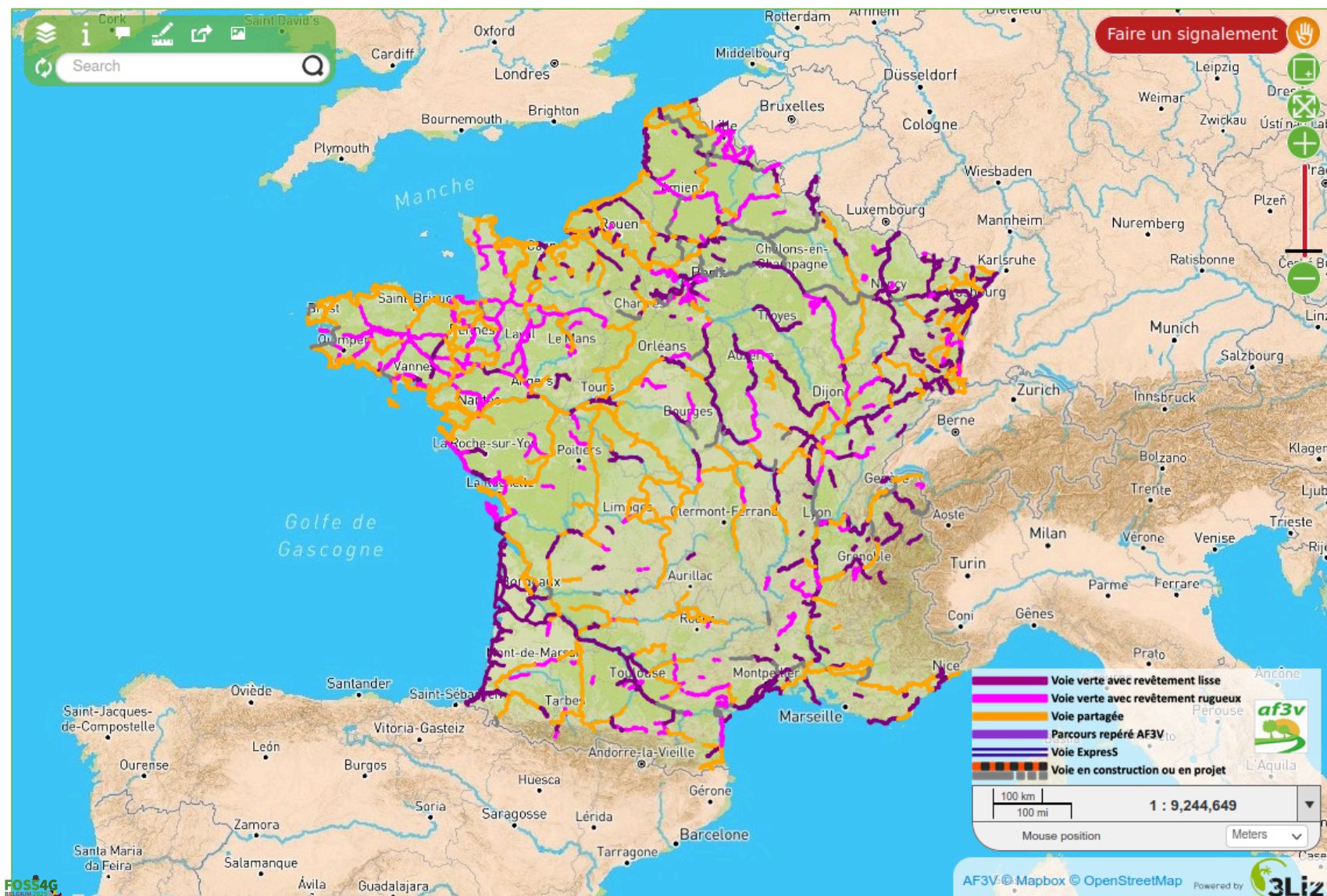
Mètres



Embed in another website - AF3V

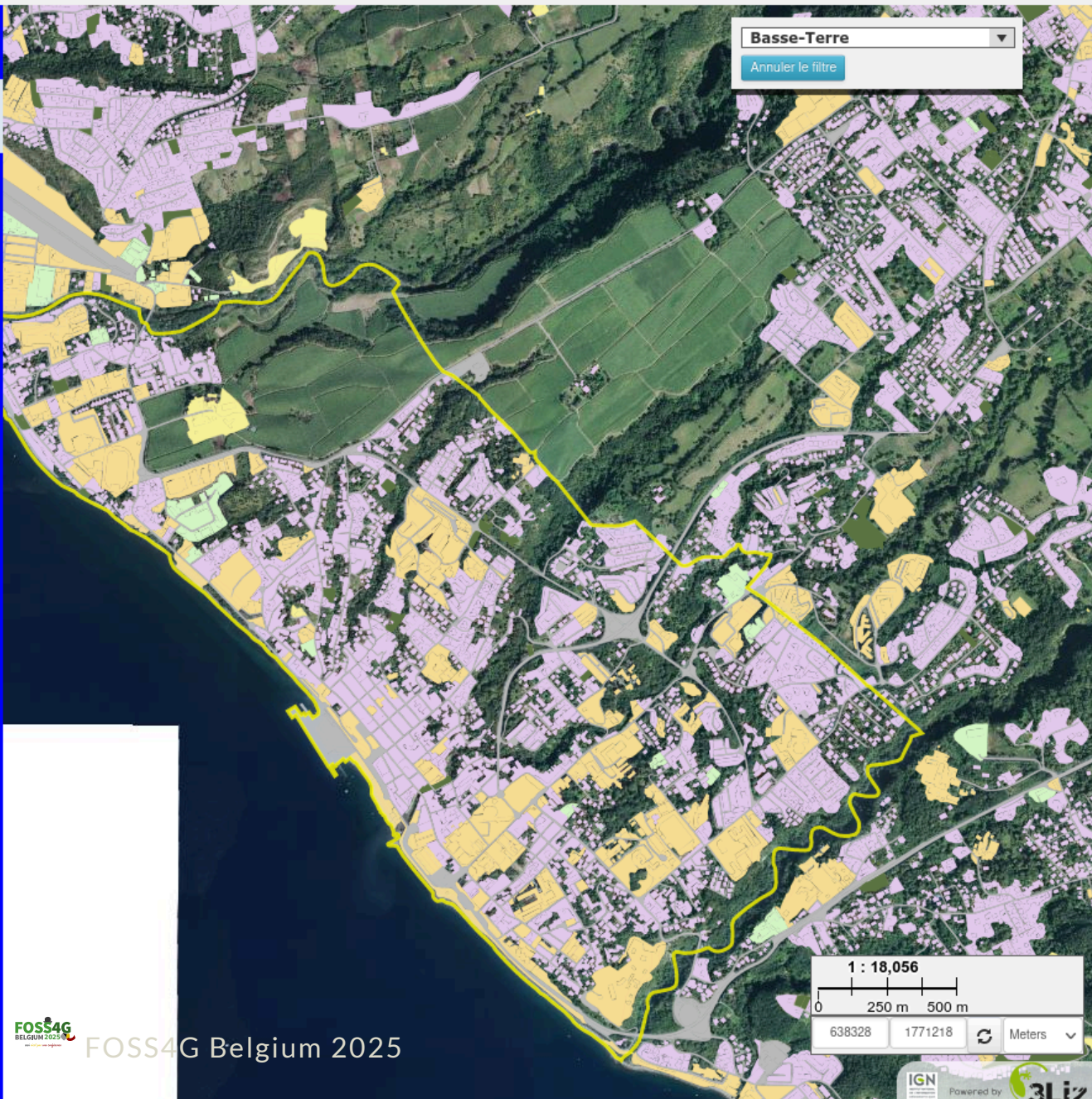
- Possible to embed Lizmap in another website
- AF3V : French association about bicycle routes
- <https://Af3v.org>

Accueil > Les Voies Vertes > Carte du schéma national des véloroutes et voies vertes



Dataviz with land use in Guadeloupe

- Guadeloupe island in the Caribbean
- Valorisation of Land Use data
- Evolution of Land use data
- https://cartophyl.lizmap.com/demo/index.php/view/map/?repository=cartophyl&project=indic_GP#



03-Artificialisation par destination

Démographie 2017

Population : 390,254 habitants

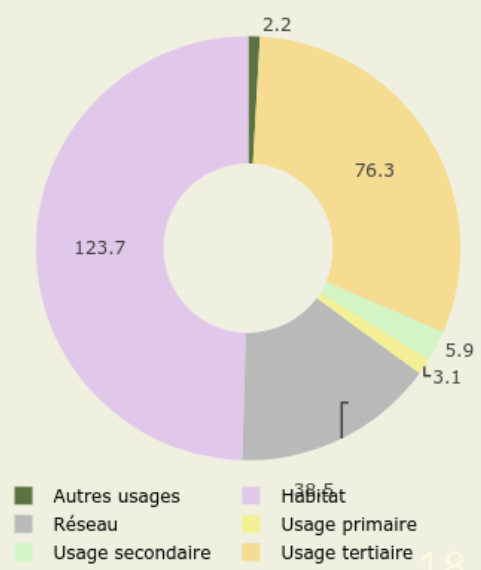
Superficie totale : 1,628.4 km²

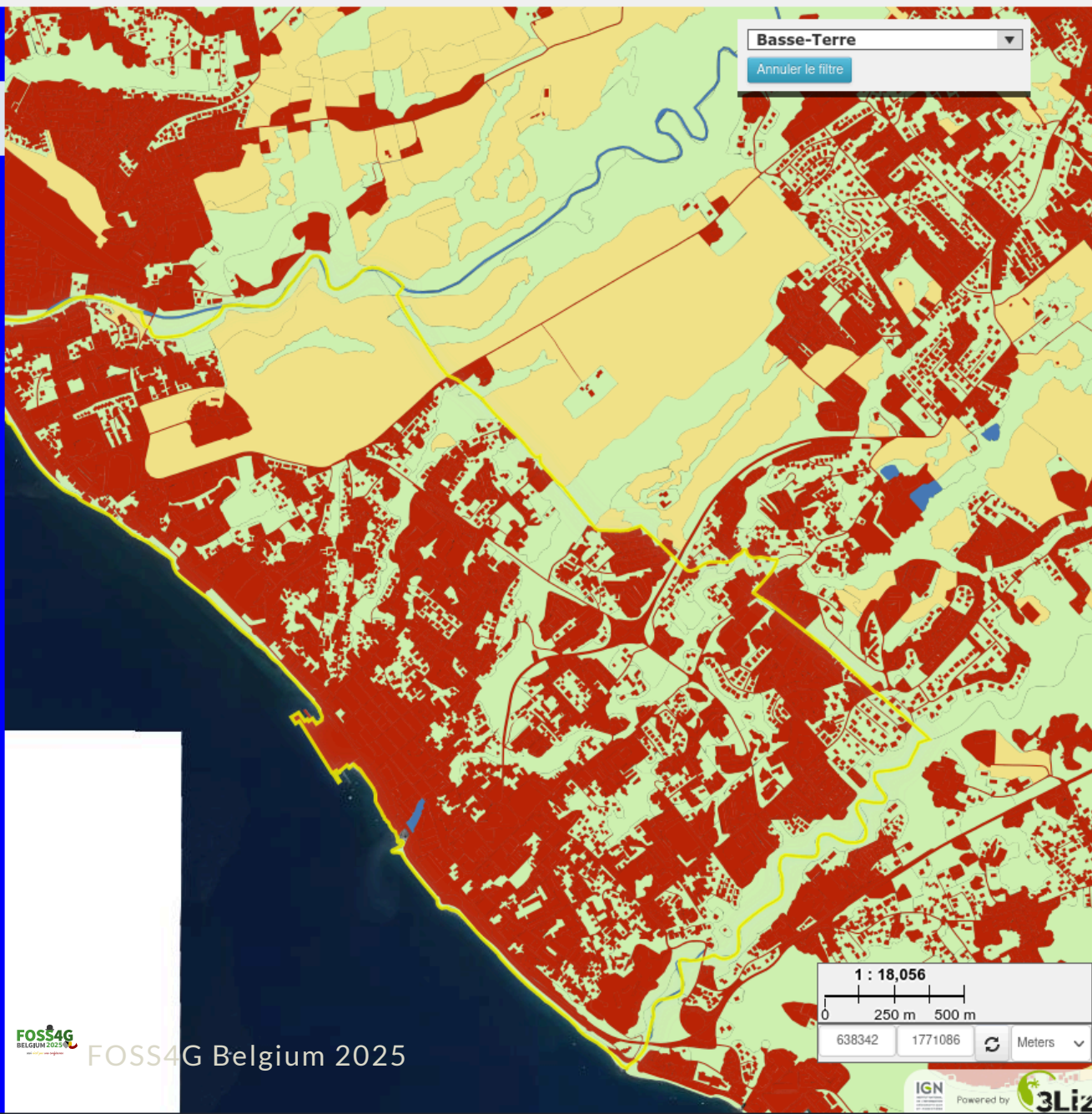
Densité : 14,506 habitants au km²

Usage des sols artificialisés en 2017

Surfaces artificialisées en fonction des usages auxquelles elles sont destinées en 2017.

L'analyse de la destination permet de déterminer quel usage des sols génère la part la plus importante d'artificialisation sur un territoire.





Basse-Terre

Annuler le filtre

01-Les grandes composantes du territoire

Démographie 2017

Population : 390,254 habitants
Superficie totale : 1,628.4 km²
Densité : 14,506 habitants au km²

Les grandes composantes du territoire en 2017

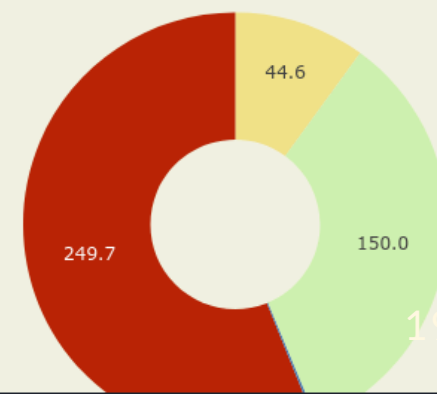
Indicateurs proposant 4 grands types d'occupation du sol sur le territoire de la Guadeloupe en 2017.

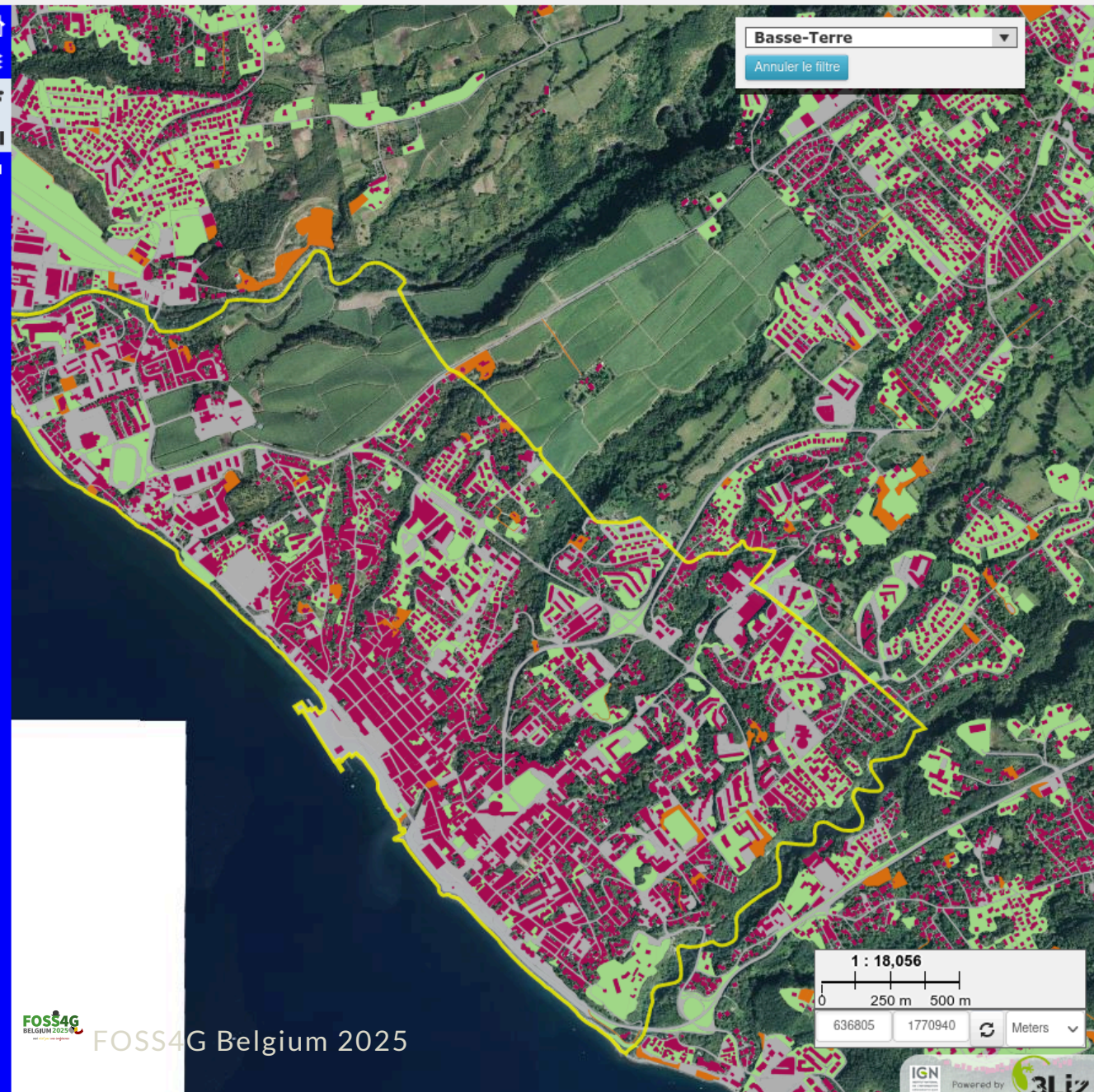
- 1) Les espaces artificialisés (*)
- 2) Les espaces agricoles (**)
- 3) Les naturels et forestiers (**)
- 4) les espaces en eau(***)

(*) : nomenclature établie par le décret n° 2022-763 du 29 avril 2022

(**) : excluant toutes composantes d'artificialisation ou surface en eau incluse dans le milieu.

(***) : ne prend pas en compte que les surfaces d'eau incluses dans la limite terrestre.





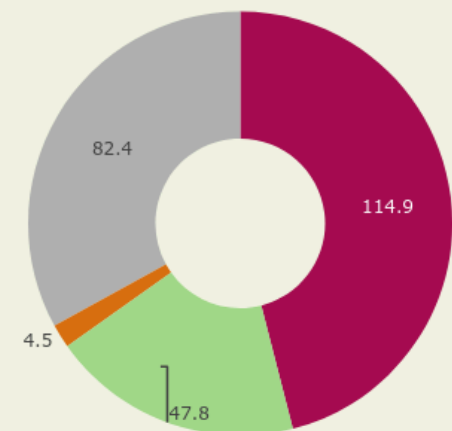
04-Couverture des sols artificialisés

Démographie 2017

Population : 390,254 habitants
Superficie totale : 1,628.4 km²
Densité : 14,506 habitants au km²

Nature des sols artificialisés

Identification de la nature des sols artificialisés à partir du niveau 2 de la nomenclature de l'OCS, à savoir s'il s'agit de sols perméables végétalisés ou non végétalisés ou de sols imperméables bâtis ou non bâtis.



- Imperméable bâti
- Imperméable non bâti
- Perméable non végétalisé
- Perméable végétalisé

Identify feature

- ADUGA, Urban planning agency
- Highlight key figures of **local business parks**
- Use core **QGIS HTML Maptip** to use **QGIS expressions**

Popul

Pôle économique

Espace industriel nord

Foncier :

488 ha de surface brute dont 434 ha à vocation économique dont :



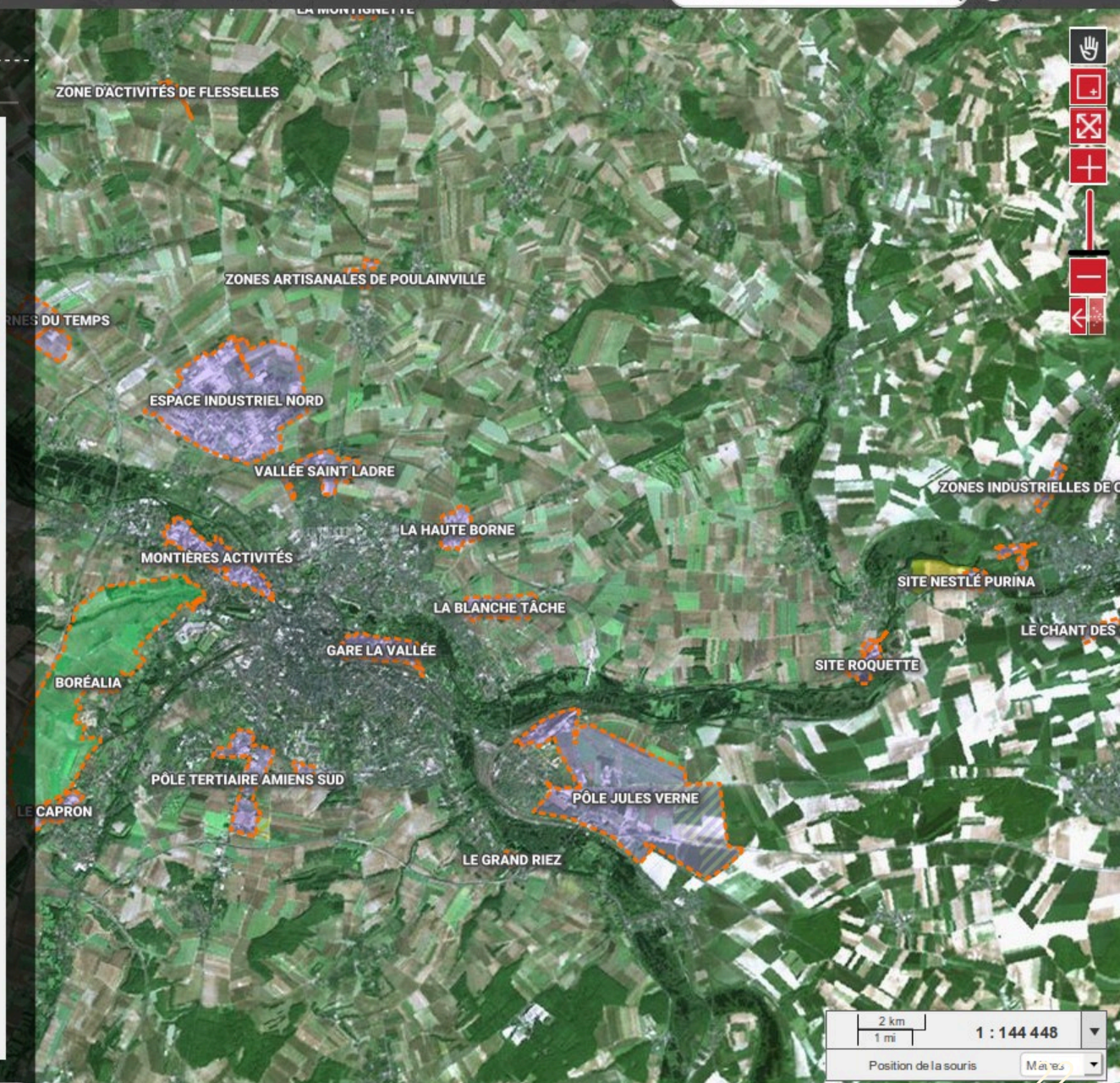
Surface disponible Surface occupée Surface en projet

Economie :

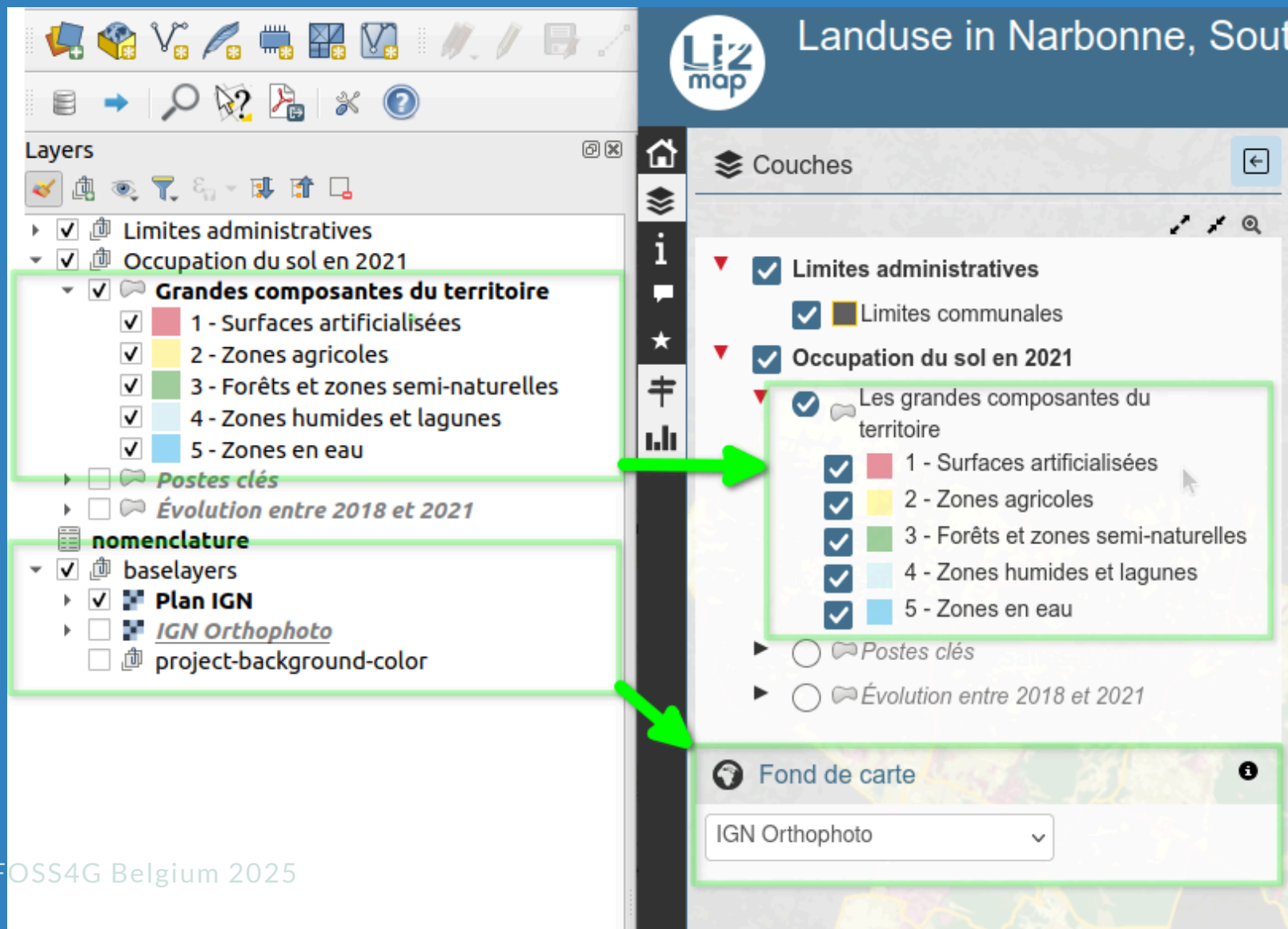
350 établissements recensés pour 10661 emplois estimés :

	Industrie	4856
	Commerce de détail et restauration	172
	BTP	884
	Transport, logistique et commerce de gros	2101
	Tertiaire supérieur	518
	Fonctions supports	577
	Services collectifs	1310
	Services de proximité	150
	Agriculture	0
	Culture, loisirs, tourisme	4

Fermer



Treewiew



The screenshot displays the Lizmap web interface for a map titled "Landuse in Narbonne, South". The interface is divided into two main panels: "Layers" on the left and "Couches" on the right.

Layers Panel (Left):

- ☒ **Limites administratives**
- ☒ **Occupation du sol en 2021**
 - ☒ **Grandes composantes du territoire**
 - ☒ 1 - Surfaces artificialisées
 - ☒ 2 - Zones agricoles
 - ☒ 3 - Forêts et zones semi-naturelles
 - ☒ 4 - Zones humides et lagunes
 - ☒ 5 - Zones en eau
 - ☐ Postes clés
 - ☐ Évolution entre 2018 et 2021
- nomenclature**
 - ☒ baselayers
 - ☒ **Plan IGN**
 - ☐ **IGN Orthophoto**
 - ☐ project-background-color

Couches Panel (Right):

- ☒ **Limites administratives**
 - ☒ Limites communales
- ☒ **Occupation du sol en 2021**
 - ☒ **Les grandes composantes du territoire**
 - ☒ 1 - Surfaces artificialisées
 - ☒ 2 - Zones agricoles
 - ☒ 3 - Forêts et zones semi-naturelles
 - ☒ 4 - Zones humides et lagunes
 - ☒ 5 - Zones en eau
 - ☐ Postes clés
 - ☐ Évolution entre 2018 et 2021
- Fond de carte**
 - IGN Orthophoto

Green boxes highlight the "Grandes composantes du territoire" section in the "Layers" panel, the "Les grandes composantes du territoire" section in the "Couches" panel, and the "baselayers" section in the "Layers" panel. Green arrows point from the "Grandes composantes du territoire" section in the "Layers" panel to the "Les grandes composantes du territoire" section in the "Couches" panel, and from the "baselayers" section in the "Layers" panel to the "Fond de carte" section in the "Couches" panel.

Attribute table

Using DataTables, a JavaScript library to display dynamic tables



Layer editing

Setting up forms in QGIS desktop can be **very** powerful !

thanks to many features such as Drag&Drop designer, container, visibility by expression, constraints, widgets...

Layer editing

Different kind of **native** editing widgets from QGIS **desktop** : date picker, dropdown, text input, checkox...

Liz map Observations faunistiques en Polynésie Cartes

Search connect

Edition Close

Edition

Manage point/vertex coordinates

Faunal observations

Observation

General

Name

Firstname

Date* Month Day Hour
Minutes

Type*

Species*

Detailed information

Sea status

Mode

Distance

☐ Single individual ?

Illustration

Picture ☐ Update

Parcourir... Aucun fichier sélectionné

After saved*

Close form

Reset Cancel Save

20 km 1 : 577 791
5 m
Mouse position Meters

Fond : ©IGN Conditions d'utilisation Powered by 3Liz

Layer editing

Constraints with a QGIS expression

Layer Properties — observations — Attributes Form

Drag and Drop Designer

Available Widgets

- Fields
 - 123 id
 - abc description
 - abc photo1
 - abc photo2
 - abc id_animal
 - date_obs
 - login
 - t/f has_details
- Relations
 - rel
- Actions
- Other Widgets
 - QML Widget
 - HTML Widget

Form Layout

- Information
 - description
 - has_details
 - id_animal
- Details
- Photos

Widget Display

- ☒ Show label

General

Alias: Description

Comment: Description

☒ Editable ☐ Reuse last entered value ☐ Label on top

Widget Type

Text Edit

☒ Multiline ☐ HTML

Constraints

☐ Not null ☐ Enforce not null constraint

☐ Unique ☐ Enforce unique constraint

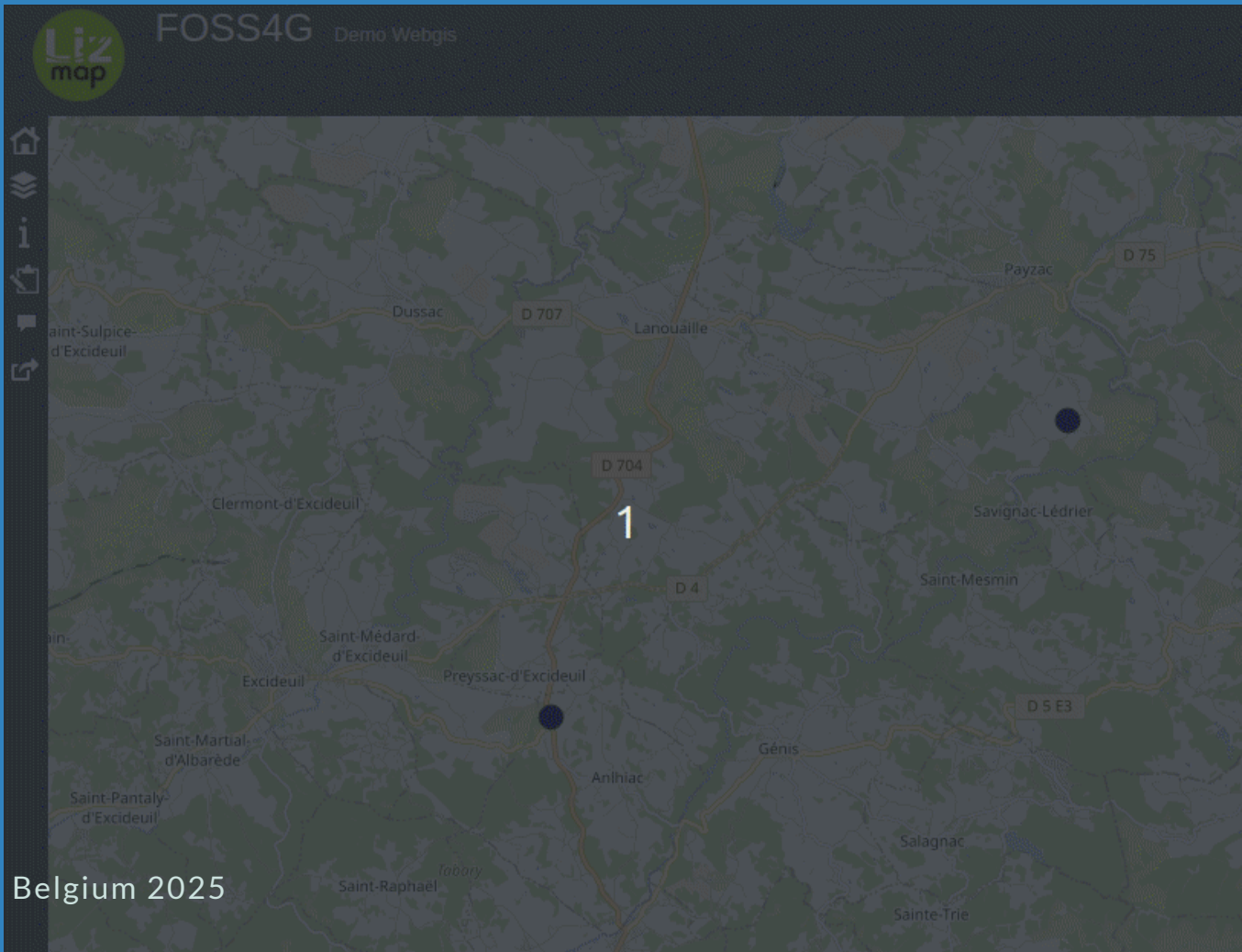
Expression: `length( "description" ) >= 30`

Expression description: Description must have at least 30 characters

Apply Cancel OK

Layer editing

Constraints with a **QGIS** expression



Layer editing - Extra capabilities

- **Geolocation**, GPS accuracy control for **field** mapping
- **Snapping** with existing features
- Geometry toolbar
 - Reverse line geometry etc

Form filtering in Narbonne

- Find local products and markets
- Using form filtering : checkbox or dropdown menu
- https://lizmap.legrandnarbonne.com/index.php/view/map/?repository=public&project=circuit_court



Cocher les cases pour chercher des points de vente.

128 résultats

Désactiver le filtre

Type de point de vente

- ☐ Aire de vente (8)
- ☐ Caveau (44)
- ☐ Magasin de produits locaux (13)
- ☐ Marché (4)
- ☐ Paniers et colis (13)
- ☐ Vente à la ferme (38)
- ☐ Autre commerce de produits locaux

Type de produits

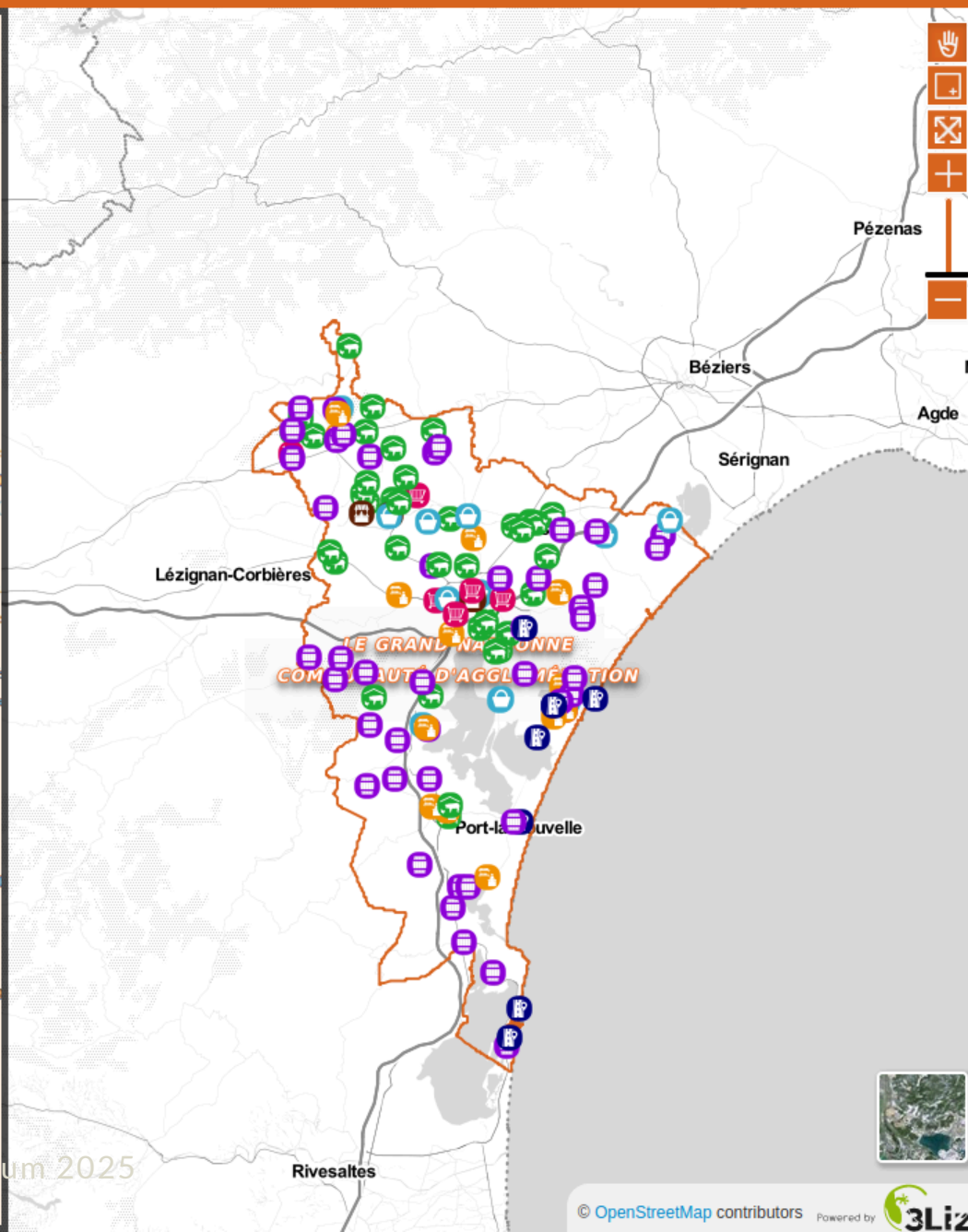
- ☐ Boissons alcoolisées (72)
- ☐ Cosmétiques et huiles essentielles
- ☐ Fromages et produits laitiers (21)
- ☐ Fruits, légumes et champignons (8)
- ☐ Horticulture (8)
- ☐ Huiles et condiments (49)
- ☐ Jus et boissons non alcoolisées (3)
- ☐ Miel, confitures et autres produits
- ☐ Pain, biscuits et céréales (27)
- ☐ Plats cuisinés et produits apéritifs
- ☐ Produits de la mer ou d'eau douce
- ☐ Viandes, volailles et oeufs (33)
- ☐ Autres (6)

Marques, mentions et labels

- ☐ Accueil Paysan (1)
- ☐ Agriculture Biologique (22)
- ☐ Agriculture Biologique en partie
- ☐ Bienvenue à la Ferme (19)
- ☐ Club Table & Vin de l'Aude (20)
- ☐ HVE (1)
- ☐ Marque Valeurs Parc Naturel Régional
- ☐ Pays Cathare (53)
- ☐ Qualité Sud de France (41)
- ☐ Qualité Tourisme (2)
- ☐ Vignobles et Découvertes (29)

Sem 4G

FOSS4G Belgium 2025



Asinerie du Rivage - Coursan

Vente à la ferme - Point de vente sur l'exploitation de produits cosmétiques au lait d'ânesse et de miel. En conversion bio.



Fiche détaillée

GAEC Le Perié - Mailhac

Vente à la ferme - Point de vente en direct de légumes et fruits frais et de saison. Vente également de produits transformés (confitures et conserves)



Fiche détaillée

Top Fruits - Argeliers

Vente à la ferme - Cueillette à la ferme de fruits et légumes frais, directement dans les champs et les vergers. A cueillir: pêches, nectarines, abricots, cerises, fraises, prunes, tomates, raisins de table, aubergines, oignons doux, courgettes, concombres.



Fiche détaillée

Les Roumanisses - Mailhac

Vente à la ferme - Vente directe d'huiles essentielles et de produits cosmétiques de l'exploitation. Vente également d'autres produits de producteurs de la région.



Fiche détaillée

Extent Lizmap core with some additional modules 🌟

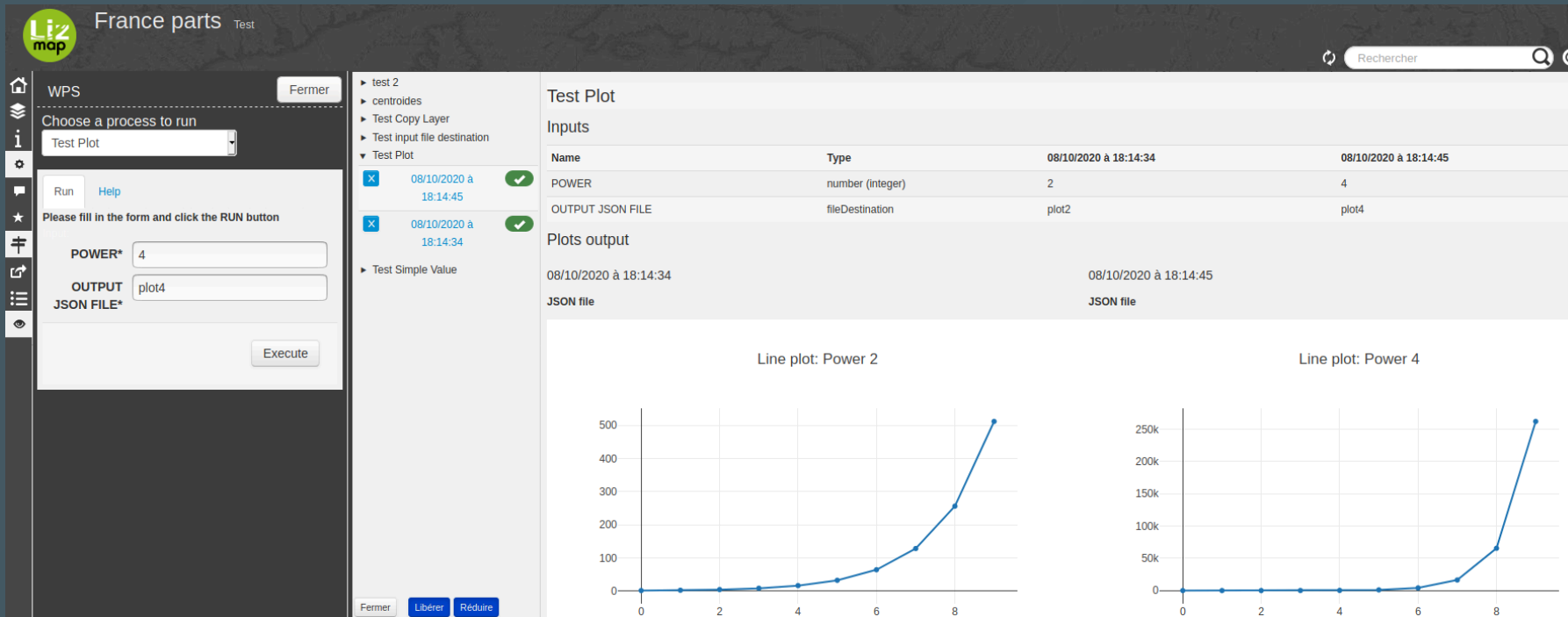
AltiProfil

- Developed by a Lizmap user in the Indian Ocean (Arnaud Vandecasteele)
- <https://github.com/arno974/lizmap-altiProfil>



Web Processing Service

- To expose QGIS Processing (the toolbox and modeler) on the web



The screenshot displays the Lizmap web interface for the 'Test Plot' process. The interface is divided into several sections:

- WPS (Web Processing Service) Panel:**
 - Header: 'France parts Test' with a 'Fermer' button.
 - Section: 'Choose a process to run' with a dropdown menu showing 'Test Plot'.
 - Buttons: 'Run' and 'Help'.
 - Text: 'Please fill in the form and click the RUN button'.
 - Form fields:
 - 'POWER*' with a value of 4.
 - 'OUTPUT JSON FILE*' with a value of 'plot4'.
 - 'Execute' button.
- Process List Panel:**
 - Tree view showing 'test 2', 'centroïdes', 'Test Copy Layer', 'Test input file destination', and 'Test Plot'.
 - Two entries for 'Test Plot' are shown with status icons (X and checkmark) and timestamps.
 - 'Test Simple Value' is also listed.
 - Buttons at the bottom: 'Fermer', 'Libérer', and 'Réduire'.
- Test Plot Details Panel:**
 - Section: 'Inputs'

Name	Type	08/10/2020 à 18:14:34	08/10/2020 à 18:14:45
POWER	number (integer)	2	4
OUTPUT JSON FILE	fileDestination	plot2	plot4
 - Section: 'Plots output'
 - Two plots are shown: 'Line plot: Power 2' and 'Line plot: Power 4'.
 - The 'Line plot: Power 2' shows a curve starting at (0,0) and ending at approximately (9, 500).
 - The 'Line plot: Power 4' shows a curve starting at (0,0) and ending at approximately (9, 250k).

Extend Lizmap with some additional JavaScript

<https://github.com/3liz/lizmap-javascript-scripts>



Lizmap JavaScript scripts

You can use those JavaScript scripts to enhance Lizmap Web Client. Documentation is [here](#).

You might have a `README.md` inside the directory with more information and a demo.

Some of the JavaScript codes presented are self-sufficient while others are examples to help you create the functionality you want.

If you are developing a new feature, do not hesitate to let us know so that we can integrate it.

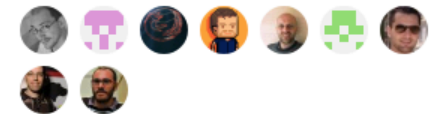
Examples of available Javascript code:

- Data:
 - [Edit GPX](#). Import/Export GPX
 - [IGN web services](#)
 - [WMTS direct access](#)
- Misc:
 - [Add keyboard shortcuts](#)
 - [Google Street View](#)
 - [Translation](#) To provide a Lizmap project in many languages
 - [Refresh some layers periodically](#)
- Tools:
 - [Attribute table column filter](#)
 - [Dynamic form field visibility](#)
 - [Multiple atlas](#), deprecated since LWC 3.4
 - [Point buffer on map](#) Display a buffer on the map
 - [Simple filter](#)
 - [Smart filter](#)
 - [Snapping while editing](#), deprecated since LWC 3.4
 - [Zoom to feature at startup and show popup](#)
- UI:
 - [Add documentation](#) with buttons and a dock
 - [Add hamburger menu](#) for a better UX on a small screen
 - [Background selector](#), like on Google Maps
 - [Group collapse](#)

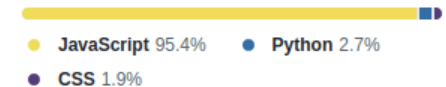
Packages

No packages published
[Publish your first package](#)

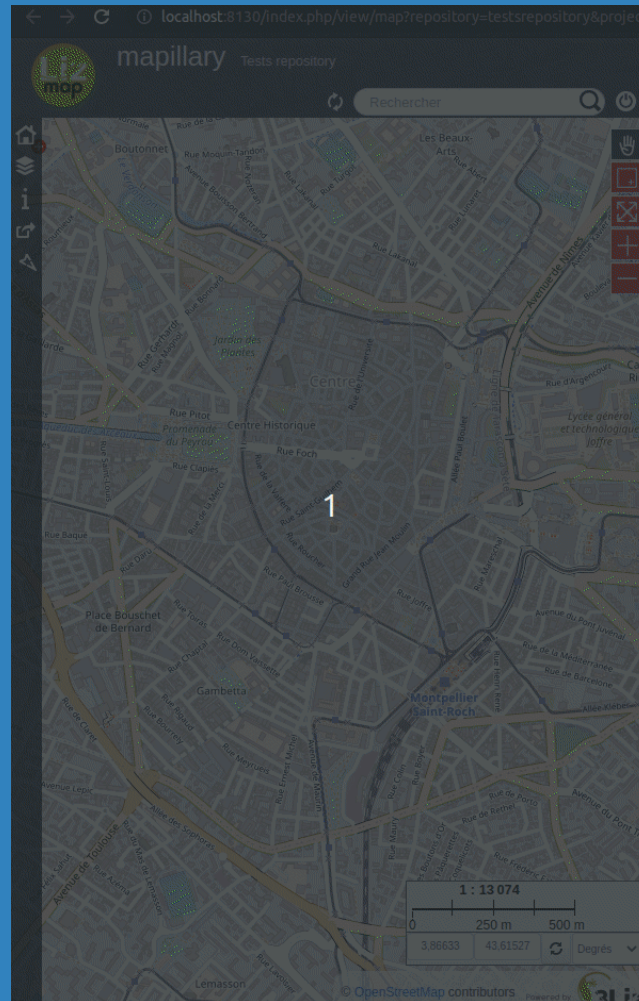
Contributors 9



Languages



Add a Mapillary viewer

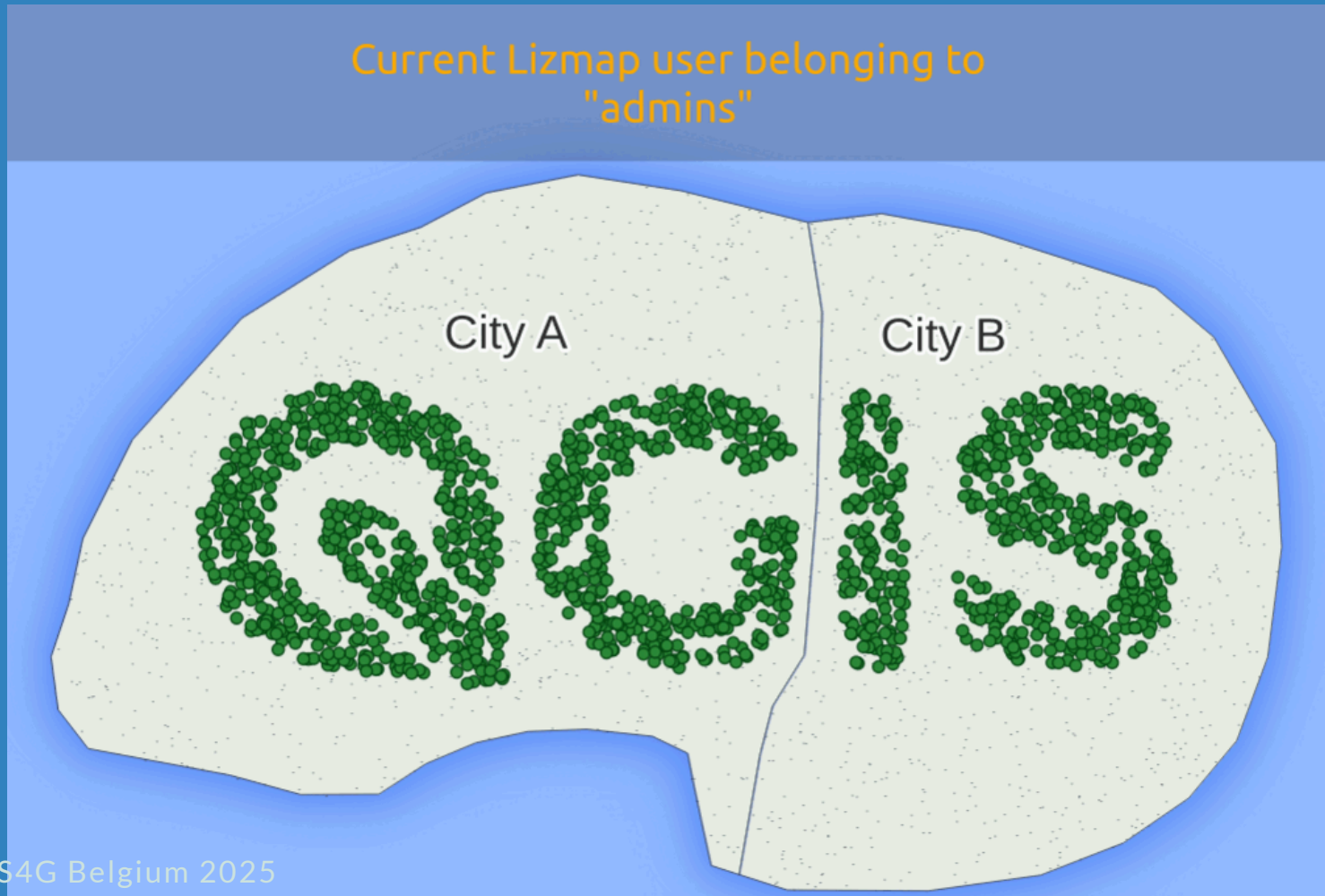


Access control

- Possible to have users and groups
- Different level of filtering and check access control :
 - Make group of projects visible or not on the landing page
 - Make projects visible or not
 - Make layers visible or not
 - Make editing available or not for a layer
 - Within the same layer, make features visible or not (either by attribute or by spatial intersection)

Access control

- Spatial filtering, like a **mask layer**



A growing community

- Different kind of users
 - Private companies
 - Public organizations
 - Research centers
- Main contribution from the **Open Source** community in
 - **translations**
 - the **Javascript** library
 - helping each other on discourse:
<https://discourse.osgeo.org/c/qgis/lizmap/48>
 - and writing some **documentation**

Translations

- <https://www.transifex.com/3liz-1/lizmap-locales/dashboard/>
- 27 languages on Transifex
 - English and French: 100%
 - Dutch: 81%



Freely usable and used worldwide

- Groundwater mapping **Western cape, South Africa**
<https://www.groundwaterinfo.africa>
- **Switzerland** <https://www.wandelderzeit.ch/>
- Faunalia (**Italia**) <https://lizmap.faunalia.eu/>
- Consorzio Toscana Nord (**Italy**) <http://geoportale.cbtoscananord.it>
- Município de Arraiolos (**Portugal**) <http://pdm-revisao.municipioarraiolos.pt/>
- SAERI (**South Atlantic**):
https://data.saeri.org/saeri_webgis/lizmap/www/



Couches

Fermer

Légende

Fauna

☐ Striated caracara potential breeders

Seabirds - Annual count of breeding pairs

☐ Gentoo penguins

☐ Southern rockhopper penguins

☐ Black-browed albatross

☐ Southern giant petrels

Island wide count of breeding pairs

☐ Black-browed albatross 2010

☐ Southern rockhopper penguins 2010

☐ Gentoo penguins 2010

☐ Southern giant petrels

Marine mammals breeding pairs count

☐ South America Fur Seal

☐ Southern Elephant Seal

☐ Southern Sea Lion

Flora

☐ Tussac islands (percentage of tussac coverage)

Population 2016

☐ Inhabited places

tourism

Cruise ships

☐ Cruise ships total passengers from 2012 to 2017

☐ Cruise ships total passengers in 2016/2017 season

FIGAS total flights and total passengers

☐ Total FIGAS passengers flying to camp

☐ Total FIGAS flights to camp destinations

FIGAS total flights per passenger category

☐ figas_total_flights_per_camp_residents

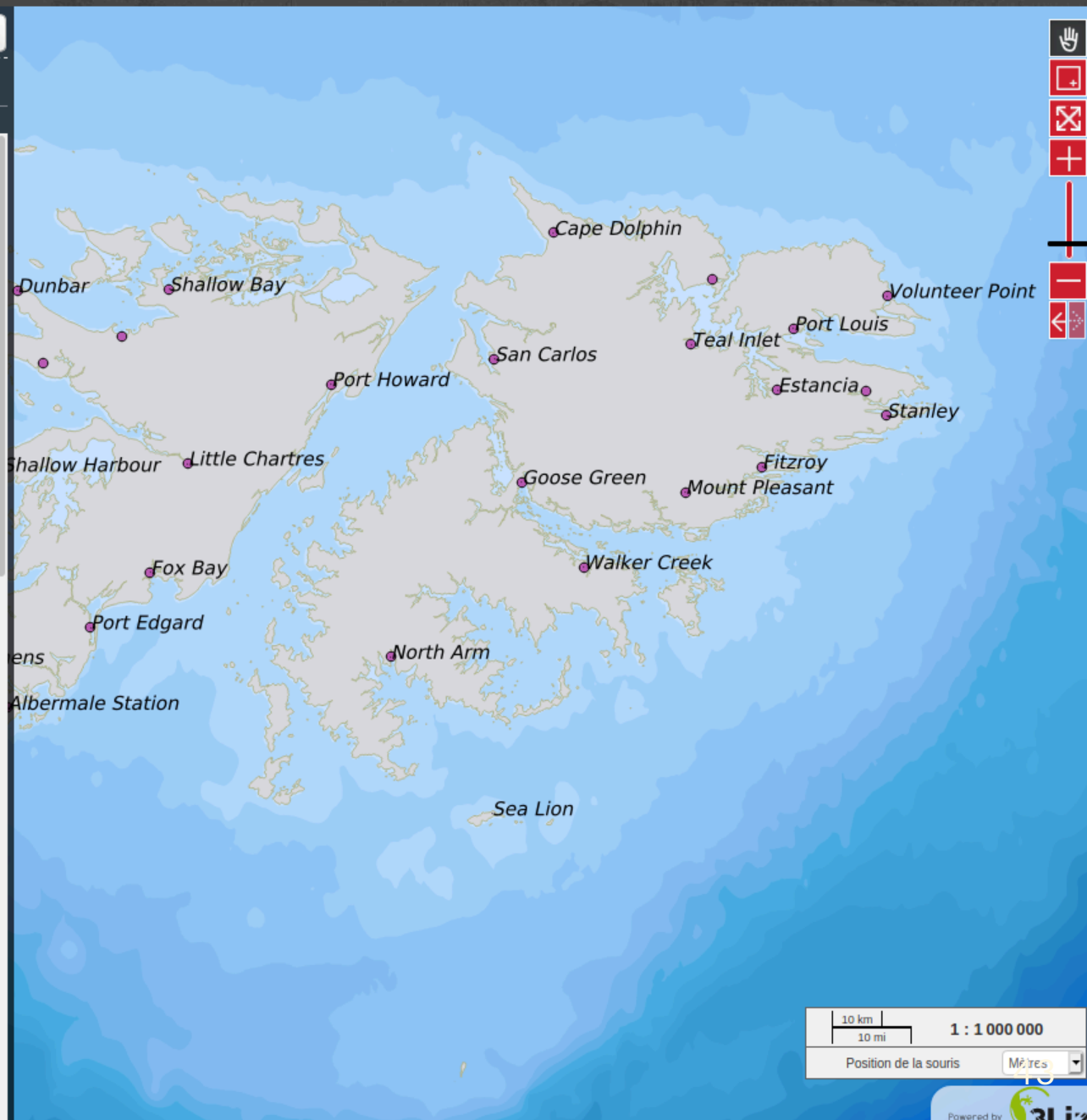
☐ figas_total_flights_per_tourists

☐ figas_total_flights_per_temporary_residents

☐ figas_total_flights_per_stanley_residents

FOSS4G

RELEASED 2025



Want to try Lizmap?

Ask us!



info@3liz.com

Questions

Thank you for your attention.

